

HealthConnect Clinic

Experience Lab Internship Programme

Week 8 Assignment

Data Science Track

Final Model Selection, Documentation & Presentation

SOGA Para

Data Science Track

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Executive Summary

This report presents the work carried out during Week 8 of the AnalystLab Africa Experience Lab programme, as part of the HealthConnect Clinic project. This week marks the final phase of the project: integration, presentation and showcase.

The main deliverable of this week is the **final No-Show prediction model**, selected after a rigorous testing and refinement process carried out during Weeks 5 to 7.

The main achievements include:

- Confirmation of the final model: Baseline (Logistic Regression)
- Optimisation of the decision threshold (0.30) $\rightarrow$ F1 = 69.45%
- Exceptional Recall of 94.23% at threshold 0.30
- Reduction of False Negatives to only 28 cases
- Validation of stability (standard deviation < 0.02)
- Absence of overfitting (Train/Test gap = 0.0014)
- HC-POD collaboration with Data Analytics

The final model achieves an F1-Score of 64.35% at the default threshold (0.50), improved to 69.45% with the optimal threshold of 0.30. The Recall increases from 66.80% to 94.23%, reducing False Negatives from 161 to 28 cases. The model is stable, free of overfitting, and performs homogeneously across all segments.

1 Week 7 → Week 8 Transition

1.1 Main Output of Week 7

The main deliverable of Week 7 was a **tested and validated candidate model** with the following characteristics:

Table 1: Candidate Model Metrics - Week 7

Metric	Score
F1-Score (threshold 0.50)	64.35%
F1-Score (threshold 0.30)	69.45%
Stability (standard deviation)	0.0186
Overfitting (gap)	0.0014
10-fold cross-validation	0.6448

1.2 Most Important Test Result

The most important result of Week 7 is the **model stability** (standard deviation = $0.0186 < 0.02$) and the **absence of overfitting** (Train/Test gap = $0.0014 < 0.05$).

1.3 Major Problem Discovered

The major problem identified is the **F1-Score slightly below the objectives**:

- Cross-validation: 0.6448 vs 0.65 (gap of -0.005)
- Generalisation: 0.6336 vs 0.65 (gap of -0.016)
- Threshold optimisation: 0.6945 vs 0.70 (gap of -0.005)

1.4 Refinement Carried Out

The main refinement was the **adjustment of the decision threshold** from 0.50 to 0.30, allowing a gain of **+5.1 F1 points** and a drastic reduction of False Negatives (161 → 28).

1.5 Component Ready

The component ready for final integration is the **No-Show prediction model** (Baseline Logistic Regression).

1.6 Elements Still to Integrate

- Business interpretation with Data Analytics
- Integration into the ML Engineering pipeline
- Final presentation

1.7 Tracks Involved

Table 2: Tracks Involved in the Final Integration

Track	Involvement
Data Analytics	KPI validation and business interpretation
ML Engineering	Integration of the model into the pipeline
Project Management	Coordination of the final presentation

1.8 Presentation Objectives

- Demonstrate individual contribution
- Explain the Data Science methodology
- Present the results and their business value
- Document the cross-track collaboration

2 Final Model

2.1 Model Selection

After a rigorous testing and comparison process, the **Baseline (Logistic Regression)** was selected as the final model.

Table 3: Comparison of Tested Models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (LR)	62.13%	62.07%	66.80%	64.35%	66.02%
Random Forest	62.34%	62.65%	65.36%	63.98%	65.56%
Gradient Boosting	61.92%	62.11%	65.57%	63.79%	66.75%
SMOTE + RF	61.08%	61.84%	62.47%	62.15%	65.60%

2.2 Justification for the Choice

The Baseline was selected for the following reasons:

1. **Best F1-Score** (64.35%): best Precision/Recall balance
2. **Best Recall** (66.80%): detects more No-Shows
3. **Best at the optimal threshold** (69.45%): adjusted performance
4. **Most stable** (standard deviation < 0.02): confirmed robustness
5. **No overfitting**: validated generalisation
6. **Interpretable**: explainable coefficients

2.3 Final Configuration

Table 4: Final Model Configuration

Parameter	Value
Algorithm	Logistic Regression
max_iter	1000
random_state	42
class_weight	balanced
penalty	l2
Deployment threshold	0.30

2.4 Final Performance

Table 5: Final Model Performance

Metric	Threshold 0.50	Threshold 0.30
Accuracy	62.13%	57.59%
Precision	62.07%	54.99%
Recall	66.80%	94.23%
F1-Score	64.35%	69.45%
ROC-AUC	66.02%	66.02%

2.5 Confusion Matrix (threshold 0.30)

Table 6: Confusion Matrix at Threshold 0.30

	Predicted Attended	Predicted No-Show	Total
Actual Attended	89 (TN)	374 (FP)	463
Actual No-Show	28 (FN)	457 (TP)	485
Total	117	831	948

3 Baseline vs Final Comparison

3.1 Performance Evolution

Table 7: Model Evolution from Week 5 to Week 8

Metric	Week 5	Week 6	Week 7	Week 8
Accuracy	62.13%	62.13%	62.13%	57.59%
Precision	62.07%	62.07%	62.07%	54.99%
Recall	66.80%	66.80%	66.80%	94.23%
F1-Score	64.35%	64.35%	64.35%	69.45%
ROC-AUC	66.02%	66.02%	66.02%	66.02%
Threshold	0.50	0.50	0.30	0.30

3.2 Interpretation

The main improvement comes from the **optimisation of the decision threshold**:

- Moving from 0.50 to 0.30 improves the F1 by **+5.1 points**
- The Recall increases from 66.80% to **94.23%** (+27.4 points)
- False Negatives are reduced from 161 to **28 cases** (-82%)
- Accuracy and Precision decrease (more False Positives)
- The ROC-AUC remains identical (66.02%)

3.3 Model Strengths

Table 8: Final Model Strengths

Strength	Description
Exceptional Recall	94.23% at threshold 0.30
Reduced False Negatives	Only 28 cases (vs 161)
Improved F1-Score	+5.1 points (64.35% → 69.45%)
Stability	Standard deviation < 0.02 across 5 random seeds
No overfitting	Train/Test gap = 0.0014
Interpretability	Explainable coefficients

3.4 Model Weaknesses

Table 9: Final Model Weaknesses

Weakness	Description
Lower Accuracy	57.59% (vs 62.13% at threshold 0.50)
Lower Precision	54.99% (vs 62.07% at threshold 0.50)
False Positives	374 cases (operational cost)
F1 below objective	0.6945 vs 0.70 (gap of -0.005)
Fictional data	Limited generalisation

4 Business Interpretation

4.1 Business Value of the Model

The final model brings significant value to HealthConnect Clinic:

Table 10: Business Value

Aspect	Impact
Reduction of No-Shows	Detects 94.23% of absences
Targeted interventions	Enables personalised reminders
Slot optimisation	Better use of resources
Patient experience	Better medical follow-up
ROI	Operational savings

4.2 Justification for the 0.30 Threshold

The choice of the 0.30 threshold is a **strategic and ethical choice**:

Table 11: Threshold Comparison

Aspect	Threshold 0.50	Threshold 0.30
Recall	66.80%	94.23%
False Negatives	161	28
False Positives	198	374
F1-Score	64.35%	69.45%

In healthcare, it is better to:

- Have more unnecessary reminders (FP) than miss at-risk patients (FN)
- Detect 94% of No-Shows rather than 67%
- Have 28 FN instead of 161

4.3 Key Features and Their Impact

Table 12: Most Important Features

Feature	Coefficient	Impact
booking_lead_days	+0.6245	Very strong
previous_no_shows	+0.2879	Strong
reminder_sent_encoded	-0.1868	Moderate
distance_to_clinic_km	+0.1714	Moderate
age	-0.1678	Moderate
gender_encoded	-0.1188	Weak
patient_reliability_score	-0.0803	Weak

4.4 Business Recommendations

1. **Use the 0.30 threshold** to maximise Recall
2. **Target patients with long booking lead times**
3. **Target patients with an absence history**
4. **Strengthen reminders** (negative coefficient = protective effect)
5. **Offer teleconsultations** for distant patients

5 Segment Performance

Table 13: Performance by Age Group (threshold 0.30)

Age group	F1	Patients	Status
18-34	0.7112	267	Good
35-49	0.7410	231	Excellent
50-64	0.6727	237	Good
65+	0.6426	213	Acceptable

Interpretation:

- The 35-49 segment has the best F1 (0.7410)
- The 65+ segment has the lowest F1 (0.6426)
- All segments exceed 0.60 → homogeneous performance

6 HC-POD Collaboration

6.1 Collaborated Track

The **Data Analytics** track was the main partner for this collaboration.

6.2 Project Dependency

Table 14: Dependency with Data Analytics

Aspect	Detail
Dependency	Validated KPIs inform modelling decisions
Justification	Insights guide feature selection

6.3 Exchanges

Table 15: Information Received and Provided

Received	Provided
Validated KPIs	Feature importance
Insights on features	Optimal thresholds
Segment analysis	Predicted probabilities

6.4 Integration Activity

The integration activity consisted of:

1. Validating the consistency between KPIs and predictions
2. Identifying gaps on certain segments
3. Adjusting the decision threshold
4. Validating features by segment

6.5 Collaboration Results

Table 16: Resulting Changes

Change	Description
Improved consistency	Better match with KPIs
Performance gain	+5.1 F1 points
Reduction of FN	161 $\rightarrow$ 28 cases
Better interpretability	Features validated by segment

6.6 Evidence

- Exchange file (Google Drive)
- Screenshot of messages
- GitHub commit
- Summary document

7 Limitations and Risks

7.1 Model Limitations

Table 17: Identified Limitations

Limitation	Description
Lower Accuracy	57.59% at threshold 0.30
Lower Precision	54.99% at threshold 0.30
False Positives	374 cases (operational cost)
F1 below objective	0.6945 vs 0.70
Data	Fictional, no external validation
Modelling	Linear, no complex interactions

7.2 Risks

Table 18: Identified Risks

Risk	Description
False Positives	374 unnecessary reminders, operational cost
False Negatives	28 cases (very low)
Drift	Change in behaviours
Deployment	Need for real validation

7.3 Mitigation

1. **Optimise hyperparameters** to improve the F1
2. **Reduce False Positives** through an adaptive threshold
3. **Validate on real data** before deployment
4. **Set up monitoring** to detect drift

8 Conclusion

8.1 Summary of Achievements

This week made it possible to:

- Confirm the final model (Baseline LR)
- Optimise the threshold to 0.30 (Recall = 94.23%)
- Reduce False Negatives to 28 cases
- Document the modelling decisions
- Collaborate with Data Analytics
- Prepare the final presentation

8.2 Contribution to the HealthConnect Project

The contribution of this week to the overall project is:

- Validation of the final model
- Optimisation of the decision threshold
- Drastic reduction of False Negatives
- Alignment with business KPIs
- Foundation for the final presentation

8.3 Learnings

This project allowed me to develop:

- Classification skills
- Feature engineering skills
- Model validation skills
- Cross-track collaboration skills

8.4 Perspectives

For a future version, it would be necessary to:

- Optimise hyperparameters
- Reduce False Positives
- Use real data
- Integrate with the ML pipeline

Appendices

Appendix A: Detailed Metrics

Table 19: Complete Classification Report (threshold 0.30)

Class	Precision	Recall	F1-Score	Support
Attended (0)	0.76	0.19	0.31	463
No-Show (1)	0.55	0.94	0.69	485
Accuracy	0.5759			
Macro Avg	0.66	0.57	0.50	948
Weighted Avg	0.65	0.58	0.51	948

Appendix B: Performance by Segment

Table 20: Performance by Age Group (threshold 0.30)

Age group	F1	Patients	Status
18-34	0.7112	267	Good
35-49	0.7410	231	Excellent
50-64	0.6727	237	Good
65+	0.6426	213	Acceptable

Appendix C: References

- AnalystLab Africa Experience Lab - Week 8 Assignment Document
- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base